Paracetamol Drug Monograph & Technical Report

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1. Chemical & Core Profile

Attribute	Specification / Value
Generic Name	Paracetamol (INN) / Acetaminophen (USAN)
Chemical Formula	$C_8H_9NO_2$
Molecular Mass	151.163 g/mol
Drug Class	Analgesic (pain reliever) and Antipyretic (fever reducer)
Mechanism of Action	Primarily inhibits prostaglandin synthesis in the central nervous system (CNS) and blocks pain impulse generation peripherally.

2. Administrative & Supply Chain Identifiers

Paracetamol is distributed globally under multiple tracking formats depending on regulatory and billing setups:

Identifier Type	Format Length	Context / Example Application
National Drug Code (NDC)	10 or 11 digits	Used in the US (e.g., 50580-449-xx for Tylenol) to track lab, product, and package sizes.
GTIN (Global Trade Item Number)	14 digits	Barcode symbology standard used for retail supply scanning and global inventory management.
GSTIN Compliance ID	15 characters	Used in jurisdictions like India on commercial tax invoices for legal movement of drug batches.

3. Clinical Guidelines & Safety Protocol

Parameters	Clinical Specifications
Standard Indications	Mild-to-moderate pain management (headache, muscle ache, arthritis) and reduction of fever.
Adult Dosage Thresholds	500 mg to 1000 mg every 4-6 hours as needed. **Strict maximum of 4000 mg (4g) in a 24-hour window** to prevent severe hepatic risk.
Contraindications	Severe hepatic impairment, active severe liver disease, or known hypersensitivity to paracetamol.
Critical Warnings	**Hepatotoxicity:** Overdose or heavy concurrent alcohol usage can deplete glutathione levels, leading to acute liver failure.

4. Sample Laboratory Quality Control (QC) Grid

Assay Test	Specification Target	Observed Level	Status
Identification (IR)	Must match reference spectrum	Conforms	PASS
Purity Assay (HPLC)	98.0% - 102.0% w/w	99.65%	PASS
Dissolution Rate	Q = 80% at 30 minutes	92.4%	PASS
4-Aminophenol Limits	Not more than 0.05% w/w	0.012%	PASS

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